The Derivation of Dupire's Equation in the Framework of Schwartz Distributions

Laurent Motais de Narbonne

October 2026

Abstract

We revisit the derivation of Dupire's equation within a rigorous framework based on Schwartz distributions, in which every object is interpreted in terms of linear forms acting on test functions. Local volatility is then no longer defined through a pointwise division, but through an equality of distributions.

1 Preliminaries: distributions

Let $\mathcal{D}(\mathbb{R}) = C_c^\infty(\mathbb{R})$ denote the space of test functions (smooth and compactly supported); see [4] for the general theory.

Definition 1. *A distribution $T \in \mathcal{D}'(\mathbb{R})$ is a continuous linear form*

$$T : \varphi \in \mathcal{D}(\mathbb{R}) \longmapsto \langle T, \varphi \rangle \in \mathbb{R}.$$

Definition 2. *The derivative of a distribution T is defined by*

$$\langle T', \varphi \rangle := -\langle T, \varphi' \rangle, \qquad \forall \varphi \in \mathcal{D}(\mathbb{R}).$$

This makes it possible to give meaning to the derivatives of non-smooth functions.

2 Option prices as distributions

Let $C(K,T)$ be the price of a European call with strike K and maturity T. We define the distribution associated with its second derivative with respect to K:

$$\langle p_T, \varphi \rangle := \left\langle \frac{\partial^2 C}{\partial K^2}(\cdot, T), \varphi \right\rangle = \int_{\mathbb{R}} \frac{\partial^2 C}{\partial K^2}(K,T)\, \varphi(K)\, dK,$$

which represents the transition density of the underlying in the sense of distributions.

By the Breeden–Litzenberger formula [1] (in the distributional sense):

$$\langle p_T, \varphi \rangle = \int_{\mathbb{R}} \varphi(K)\, p(K,T)\, dK.$$

Here, p_T is interpreted as a distribution, which may contain Dirac masses or singularities.

3 The Fokker–Planck equation in terms of linear forms

Assume that the underlying S_t follows diffusion-type dynamics:

$$dS_t = \sigma_{\text{loc}}(S_t, t)\, S_t\, dW_t.$$

The density $p(K,T)$ satisfies the Fokker–Planck equation:

$$\frac{\partial p}{\partial T} = \frac{1}{2}\frac{\partial^2}{\partial K^2}\Big(\sigma^2_{\text{loc}}(K,T)\, K^2\, p(K,T)\Big),$$

which must be understood in the sense of distributions:

$$\left\langle \frac{\partial p_T}{\partial T}, \varphi \right\rangle = \frac{1}{2}\left\langle \frac{\partial^2}{\partial K^2}\Big(\sigma^2_{\text{loc}}(\cdot,T)\,(\cdot)^2\, p_T\Big), \varphi \right\rangle, \qquad \forall \varphi \in \mathcal{D}(\mathbb{R}).$$

Applying the definition of the distributional derivative twice, we obtain, for any smooth function f:

$$\left\langle \frac{\partial^2}{\partial K^2}(f\, p_T), \varphi \right\rangle = \langle f\, p_T, \varphi'' \rangle = \langle p_T, f\, \varphi'' \rangle.$$

The equation can therefore be rewritten in a fully linear form:

$$\left\langle \frac{\partial p_T}{\partial T}, \varphi \right\rangle = \frac{1}{2}\left\langle p_T, \sigma^2_{\text{loc}}(\cdot,T)\, K^2\, \varphi'' \right\rangle.$$

This is the exact formulation of Dupire's equation [2] in the sense of distributions.

4 Expression of local volatility

In practice, Dupire writes [2] (see also [3]):

$$\sigma^2_{\text{loc}}(K,T) = \frac{\dfrac{\partial C}{\partial T}(K,T)}{\dfrac{1}{2}K^2\dfrac{\partial^2 C}{\partial K^2}(K,T)}.$$

In the distributional sense, we consider evaluations against a test function φ:

$$\left\langle \sigma^2_{\text{loc}}(\cdot,T)\, K^2\, p_T, \varphi \right\rangle = 2\left\langle \frac{\partial C}{\partial T}(\cdot,T), \varphi \right\rangle.$$

This formulation means that local volatility is defined as a distribution: rather than dividing pointwise, σ^2_{loc} is defined by the equality of distributions over all test functions.

5 Advantages of the distributional framework

1. **Singularities.** Points where p_T carries a Dirac mass (for instance an "atomic" strike) are automatically taken into account.

2. **Mathematical rigour.** Dupire's equation is well defined even when $C(K,T)$ is not of class C^2.

3. **Weak solution.** The distributional framework is equivalent to considering weak solutions of the Fokker–Planck equation.

References

[1] D. T. Breeden and R. H. Litzenberger, Prices of state-contingent claims implicit in option prices, *Journal of Business*, 51(4):621–651, 1978.

[2] B. Dupire, Pricing with a smile, *Risk*, 7(1):18–20, 1994.

[3] J. Gatheral, *The Volatility Surface: A Practitioner's Guide*, Wiley, Hoboken, NJ, 2006.

[4] L. Schwartz, *Théorie des distributions*, Hermann, Paris, 1950–1951; new edition 1966.